

Neurological and Thermodynamic Mechanisms Underlying Anomalous Presence Perception

Abstract

This hypothesis explores the neurological and thermodynamic mechanisms underlying anomalous presence perception, the experience of sensing a deceased person's presence. Among the most universally documented phenomena in human history, reported across all cultures, all eras, and all psychological profiles, it has never received a unified scientific explanation. This hypothesis proposes that anomalous presence perception is not supernatural, not pathological, and not imagined, but a precise biological consequence of five interacting mechanisms: infrasound induced cortical arousal operating below conscious sensory threshold, predictive processing failure during bereavement, Neural Persistence, the continued firing of long term potentiated neural pathways built around a specific individual, Cardiac Neural Persistence, the encoding of relational memory in the heart's own intrinsic nervous system generating somatic presence independently of cortical reconstruction, and the Negativity Threshold, a proposed residual signature left by sustained self hate, regret, and chronic stress that a visitor's subconscious can detect as mismatch even without any personal relationship to the deceased. Version 3 extends the framework in three directions: the Negativity Threshold mechanism itself, aimed at stranger reported experiences with no personal relational encoding; a distinction between actionable and unactionable threat, proposing that the disorientation and loss of reality testing reported during presence experiences comes from the absence of a defensive action available to the subconscious, not from the intensity of the threat response itself; and a resolution of the apparent conflict between predictive hallucination and genuine anomalous detection, arguing both describe the same underlying mechanism viewed from either side of it, with cultural context determining only the specific form the detected anomaly takes. The subconscious brain, trained over years to expect a person's voice, scent, and energetic signature, does not accept their absence. It generates them instead. And the body remembers too. This hypothesis draws on thermodynamics, neuroscience, olfactory biology, cardiac physiology, relational self theory, and the quantum observer analogy to argue that what humanity calls paranormal experience is the brain's and the body's most profound act of loyalty to those it has loved, and, in its extended form, its most immediate act of self protection against what it has flagged as unsafe. When someone dies, we do not only lose them, we lose the version of ourselves that could only exist in their presence. The hypothesis remains strongest in explaining presence perception following the death of a loved one or close relation, where neural persistence, cardiac conditioning, and relational self dissolution converge simultaneously, and is now more developed, though still evolving, in accounting for stranger reported experiences at culturally designated haunted or suicide associated locations, through the addition of the Negativity Threshold mechanism.

1. Background and Motivation

Human beings have reported experiences of sensing the presence of deceased individuals across all known cultures and throughout recorded history. These experiences range from subtle feelings of

familiarity in specific locations to vivid perceptions of another person's presence. Modern science has largely dismissed these as superstition or hallucination without proposing adequate mechanistic explanations for why they occur so universally and consistently.

This hypothesis takes a different position: that paranormal experiences are real neurological events with identifiable physical and psychological causes. Understanding them scientifically does not diminish their significance — it deepens it.

2. Core Hypothesis Statement

Paranormal experiences — specifically the sensing of a deceased person's presence — arise from a convergence of five distinct but interacting mechanisms:

1. **Thermodynamic energy transformation:** Human body energy does not disappear at death but transforms into heat, chemical, and environmental energy that continues to exist in the immediate environment.
2. **Subconscious associative memory reconstruction:** The brain builds detailed neural maps of loved ones during life. After death, these maps remain active and reconstruct a sense of presence when triggered by environmental cues associated with the deceased.
3. **Undetected physical stimuli:** Infrasound below 20Hz, electromagnetic field fluctuations, and decomposition-related chemical compounds — all operating outside normal human sensory detection ranges — trigger neurological responses misinterpreted as supernatural presence.
4. **Predictive processing and cultural priming:** The brain continuously generates predictions about its environment. Prior exposure to cultural narratives about ghosts, death, and haunted spaces primes the brain to interpret ambiguous stimuli as presence-related experiences.
5. **Cardiac neural persistence:** The heart's intrinsic nervous system encodes relational memory through conditioned heart rate variability patterns built around specific individuals. After death, these cardiac pathways persist and fire without their expected stimulus — generating somatic presence independently of cortical reconstruction, and responding to emotionally significant stimuli before conscious brain processing occurs.

3. Thermodynamic Foundation

The First Law of Thermodynamics states that energy cannot be created or destroyed, only transformed from one form to another. The human body contains approximately 80,000 kJ of stored chemical energy, alongside significant thermal, electrical, and kinetic energy generated continuously by metabolic processes. At the moment of death, this energy does not vanish — it begins a sequence of transformations. Body heat — maintained at 37°C throughout life — dissipates into the surrounding environment within hours, measurably raising the ambient temperature of a closed room. Chemical energy stored in cellular structures releases gradually through decomposition, powering microbial activity that continues for months. The electrical energy of the nervous system — action potentials, synaptic transmission — ceases as an organised system but its constituent ionic charges redistribute into the surrounding fluid environment. None of this energy disappears. It changes form. The Second Law of Thermodynamics adds a critical dimension: energy transformation

is directional. Energy moves from concentrated, organised states toward diffuse, disordered states — a process called entropy. A living human being represents an extraordinarily concentrated, organised energy system. Death marks the beginning of entropy — the gradual dispersal of that concentrated energy outward into the environment.

The space a person inhabited for years accumulates trace evidence of this dispersal: residual thermal patterns in walls and furniture, volatile organic compounds released by skin cells embedded in fabric, electromagnetic traces in the electromagnetic field of a room shaped over time by the body's bioelectric activity. These are not supernatural emissions. They are the physical consequences of the Second Law — measurable, real, and potentially detectable by a subconscious brain trained over years to recognise the specific energy signature of that individual.

This transformed energy continues to exist in the environment, particularly in spaces where a person spent significant time. The hypothesis proposes that a subconscious brain deeply familiar with a specific person's energy signature — built through years of close association — may retain sensitivity to these transformed energy patterns in ways not yet fully understood.

4. Sensory Limitation and Hidden Stimuli

Human sensory perception covers only a fraction of existing physical phenomena:

- Vision: 430–770 THz (visible light spectrum only)
- Hearing: 20 Hz – 20 kHz (audible sound range only)
- Electromagnetic fields: Not consciously detectable by humans

Outside these ranges, significant physical activity continues undetected. Infrasound — sound below 20 Hz — is generated by decomposition processes, wind movement through structures, industrial equipment, and underground water movement. Research has demonstrated that infrasound exposure produces anxiety, unease, peripheral visual disturbances, and critically — a sensation of an unseen presence. This effect was formally documented by Vic Tandy and Tony Lawrence in their 1998 paper 'The Ghost in the Machine,' published in the Journal of the Society for Psychical Research — the first scientific account directly linking infrasound exposure to paranormal-type sensory experiences.

However, sensory limitation does not imply cognitive limitation. Evidence from subliminal perception research and infant olfactory studies suggests the subconscious brain may process stimuli below conscious detection thresholds. The sensation of paranormal presence may therefore represent subconscious signal processing — real stimuli detected below conscious awareness — surfacing as unexplained emotional or physical responses such as unease, goosebumps, or the feeling of being watched. A compelling parallel exists in the phenomenon of gaze detection — known in parapsychological literature as *scopaesthesia*. The human brain possesses a dedicated neural region, the Superior Temporal Sulcus (STS), specifically evolved to detect the directional orientation of another person's eyes and attention. Gaze detection and anomalous presence perception share the identical underlying architecture: environmental micro-signals processed below conscious threshold, surfaced as a specific felt experience. Both experiences are equally real. Both emerge from the same system.

"What we call paranormal is simply the mind's extraordinary ability to generate presence from memory, expectation and the physics we cannot yet fully measure."

5. Neurological Mechanisms

5.1 Default Mode Network and Grief Hallucination

When the brain is not focused on external tasks, it activates the Default Mode Network (DMN) — an internal processing state characterized by memory replay, social simulation, and self-referential thinking. In states of grief, bereavement, or relaxation in familiar environments, the DMN reconstructs detailed simulations of deceased individuals using stored neural maps. Research indicates that approximately 80% of widowed individuals report sensing the presence of their deceased spouse within the first year of bereavement. These experiences occur in individuals with no prior history of hallucination and are neurologically distinct from pathological hallucination.

5.2 Developmental Decline of Subconscious Sensitivity

This hypothesis proposes that subconscious sensory sensitivity follows a developmental arc across the human lifespan. In infancy and early childhood, the conscious mind is underdeveloped — the subconscious dominates perception entirely. Infants demonstrate this clearly: a newborn cannot consciously identify its mother, yet responds immediately and distinctly to her scent, warmth, and energetic presence — processing signals the conscious mind cannot yet access or articulate. This subconscious sensitivity is not primitive — it is extraordinarily refined.

As the conscious mind develops through childhood and adolescence, it progressively dominates perception — filtering, categorising, and rationalising incoming stimuli. Subconscious sensitivity does not disappear but becomes increasingly suppressed. This developmental suppression may explain why developmental research consistently suggests children report paranormal and fantasy experiences at higher rates than adults. Critically, this hypothesis proposes that imaginary friends in isolated children may not represent pure imagination — they may represent the subconscious mind actively seeking social and energetic connection that the real world has failed to provide.

Notably, this arc may reverse in old age. The Frontal Aging Hypothesis demonstrates that prefrontal cortex volume decreases at approximately 5% per decade from middle age onward, with corresponding decline in executive inhibition. As the brain's primary realityfiltering system weakens, subconscious processing re-emerges relatively stronger. Furthermore, elderly individuals have accumulated decades of loss — their Default Mode Network is saturated with neural maps of deceased individuals. The filter is thinnest precisely when the material available for reconstruction is greatest.

5.2.1 Neurodegeneration and the Two-Failure Model

This hypothesis extends the developmental arc to account for a distinct and more severe category of anomalous presence experience observed in neurodegenerative disease — specifically Alzheimer's disease and Lewy body dementia. These conditions produce anomalous presence perception through a fundamentally different mechanism than normal aging: not merely weakened filtering, but a simultaneous failure of both signal generation and signal verification.

In neurodegeneration, progressive synaptic loss and neuronal death produce erratic, unpredictable firing patterns across degraded neural networks. Neurons that previously encoded presence-maps of specific individuals begin firing without appropriate external input — pathological false-signal

generation. Simultaneously, the prefrontal cortex degrades at an accelerated rate, removing the mechanism that would flag these signals as internally generated. The result is a two-part failure:

1. Signal generation failure: Degraded neurons fire erratically along presence-encoded pathways, generating false presence signals without environmental stimulus.
2. Signal verification failure: The prefrontal reality-checking mechanism is simultaneously too degraded to flag these signals as internally generated.

Visual and auditory hallucinations occur in approximately 40–50% of Alzheimer's patients and in up to 75–80% of Lewy body dementia patients, significantly exceeding rates in age-matched controls. This hypothesis proposes these hallucinations are patterned outputs shaped by the neural maps the brain spent decades constructing — the presence-maps of deceased individuals encoded through long-term potentiation. The faces that appear are not arbitrary: they are the people the brain considered most important.

Testable prediction: Alzheimer's and Lewy body dementia patients should show statistically higher rates of anomalous presence perception specifically involving deceased individuals — not living strangers — compared to age-matched controls without neurodegeneration. Furthermore, the specific individuals perceived should correlate with the depth and duration of the patient's prior relationship with those individuals.

Complete Developmental Arc

Infant: No conscious mind. Subconscious fully dominant. Zero self-awareness. Maximum sensitivity.

Child: Conscious developing. Subconscious still strong. Self-awareness emerging. Imaginary friends possible.

Adult: Conscious fully dominant. Subconscious suppressed. Self-awareness complete. Minimum sensitivity.

Elderly: Conscious degrading. Brain reroutes to subconscious via neurological compensation. Neural maps of deceased richest. Maximum sensitivity returns.

Neurodegenerative (Alzheimer's / Lewy body): Two-failure model — erratic false-signal generation combined with failed verification. Presence felt as externally real without environmental cause. Pathological maximum.

5.3 Associative Memory and Environmental Triggers

Electrical stimulation of the temporal lobe produces experiences remarkably similar to reported paranormal encounters: sensing an unseen presence, hearing deceased individuals speak, feeling watched, and experiencing profound religious or spiritual states. Dr. Michael A. Persinger's God Helmet experiments at Laurentian University demonstrated that weak magnetic fields targeted at the temporal lobes reliably produced a sensed presence in the room — experienced by up to 80% of subjects as a deceased person, angel, or undefined sentient being. Persinger's proposed mechanism — interhemispheric disruption causing the brain's internal representation of another person to be externalised — is consistent with the Neural Persistence framework proposed in Section 9.

5.4 Predictive Processing and Cultural Priming

The brain operates as a predictive machine, continuously generating expectations about incoming sensory data — a framework formalised by neuroscientist Karl Friston through the Free Energy Principle, and extended to conscious perception by Anil Seth at the Sussex Centre for Consciousness Science. Prior exposure to cultural narratives — horror media, ghost stories, religious frameworks — programs these predictions. When entering environments culturally designated as death-associated, the brain pre-activates presencedetection systems, dramatically lowering the threshold for interpreting ambiguous stimuli as evidence of supernatural presence.

6. Application to Haunted Locations

The hypothesis extends naturally to explain collective paranormal experiences in specific locations. Haunted locations typically share several characteristics: historical association with death or trauma, old structural features generating infrasound, electromagnetic anomalies from aged electrical systems, and strong cultural designation as paranormal sites. Strangers entering these locations without personal connection to deceased individuals may still experience presence-sensations through the combined effect of infrasound-induced temporal lobe activation, cultural priming via predictive processing, and ambient electromagnetic fluctuations detected subconsciously.

Empirical data on documented haunted locations challenges the common assumption that sites of high death volume — hospitals, graveyards — dominate paranormal reports. Counterintuitively, the majority of reported haunted locations are educational establishments, with residential houses, hotels, and historic structures collectively accounting for far more reports than cemeteries or hospitals. Graveyards, despite their overwhelming cultural association with death and the supernatural, represent a surprisingly small fraction of documented sites. Hospitals rank even lower.

This distribution is not random. It is mechanistically meaningful. What schools, old residential buildings, and historic structures share is not death volume — it is sustained habitation. The same individuals occupying the same spaces over years, building deep relational and neural maps of those environments. This is precisely what the neural persistence framework predicts: haunting is driven not by how many people died in a place, but by how deeply and how long specific presences were encoded there. A school attended by the same students and staff for decades generates far richer conditions for residual presence than a hospital where patients cycle through in days. The data, counterintuitive as it appears, is a direct empirical prediction of this hypothesis.

This hypothesis proposes that the body's electromagnetic and thermal signatures — generated continuously over years of habitation — may leave measurable traces in the physical environment of frequently inhabited spaces. If the subconscious brain is sensitive to energy patterns below conscious threshold, as proposed in Section 4, it may register these residual traces and interpret them as the familiar presence of the individual who generated them. This mechanism remains speculative and represents a primary area requiring empirical investigation.

7. Conscious-Subconscious Integration in Anomalous

Presence Perception

Anomalous presence experience emerges from the simultaneous integration of both conscious and subconscious systems, where conscious context determines the interpretation of subconscious detection. The physical stimulus can be identical across environments — what changes is the conscious frame applied to it. Near a bonfire, thermal energy is explained and categorised. In a graveyard, identical stimuli integrate with a death context — producing presence sensation and anomalous experience.

This model addresses the specificity problem in hallucination research. Methane produced by anaerobic decomposition crosses the blood-brain barrier and acts as a competitive antagonist at NMDA receptors — weakening the brain's ability to tag its own internal simulations as imaginary. The DMN continues generating simulations of the deceased, but NMDA suppression removes the internal marker that would normally flag them as memory rather than present reality. The methane provides the intensity. The conscious narrative provides the content. Together they produce a specific, coherent hallucination — not random noise.

This model can be expressed as: Conscious Context + Subconscious Detection = Presence Experience. What critics call confirmation bias, this hypothesis identifies as mechanism.

This also resolves an apparent either/or that the hypothesis does not need to choose between. Whether the brain is simply complex enough to construct convincing hallucination on its own, or whether it is complex enough to genuinely detect something anomalous in the environment, both possibilities run through the same equation above and are not in competition. Strip away every cultural label and what remains is one detection system, picking up an ambiguous signal below conscious threshold, that a conscious mind then completes using whatever categories its era has made available to it. An early human with no religious framework and no concept of spirits has essentially one category available for an unexplained anomaly, predator. A person in a culture that has spent centuries building out ghosts, spirits, gods, and the dead, has somewhere else for that same ambiguous signal to land. The detection event does not change. Only the face the conscious mind puts on it changes. This does not settle whether something is actually being sensed in the environment or whether the brain is generating the entire experience unassisted. It only establishes that the two explanations are not rivals. They are the same mechanism, described from either side of it.

8. The Quantum Observer Parallel — Consciousness as Collapse Mechanism

A foundational insight from quantum physics reframes the entire question of what perception is. At the atomic level, no two pieces of matter ever make direct contact. What we experience as touch is not contact — it is electromagnetic repulsion between electron clouds. The sensation of touching another person's hand, feeling their warmth, sensing their physical presence — none of it is direct access to reality. All of it is a neural construction: the brain interpreting electromagnetic force data and generating the felt experience of contact. Every sensory experience a human being has ever had is a model, not a recording. The brain has never touched the world directly. It has only ever touched its own representations of it.

This has a profound implication for anomalous presence perception. If real touch is already a neural construction — if the brain generates the experience of contact from electromagnetic signals it never directly accesses — then generating the feeling of a presence from residual energy patterns, neural persistence firing, and cardiac conditioning is not an extraordinary leap. It is the brain doing exactly what it always does. The same mechanism. The same architecture. Just operating in the absence of the living person who originally generated the signal. The paranormal experience is not the brain malfunctioning. It is the brain functioning precisely as it always has — constructing experience from incomplete data — in a situation where the primary source of that data is gone. As neuroscientist Anil Seth frames it: all conscious experience is a controlled hallucination. The only difference between perceiving a

person who is present and sensing a presence who is not is the strength and source of the incoming signal. The brain does not know the difference. It never did. That is not a bug. That is the whole architecture.

A structural parallel exists between the mechanism proposed in this hypothesis and the observer effect in quantum mechanics. In quantum mechanics, a photon without observation exists as a probability wave. The moment observation occurs, the wave function collapses into a fixed particle reality. The mechanism of this collapse remains one of the deepest unsolved problems in modern physics.

In presence perception: ambiguous environmental stimuli exist as interpretive probability. The subconscious detects them below threshold but produces no specific experience. Then the conscious mind engages — and the ambiguous stimuli collapse into a specific felt presence. Both systems move from undefined to defined through the act of observation. Both leave the same question unanswered: what exactly is it about conscious observation that forces the collapse?

This parallel is presented as a structural analogy rather than a literal claim. The value is conceptual: both phenomena reveal consciousness performing the same fundamental operation across vastly different scales — collapsing undefined possibility into specific experienced reality.

9. Neural Persistence — Why The Brain Cannot Accept Absence

In 2022, Cortical Labs demonstrated that 292,000 human neurons cultured on a silicon chip could learn to play Pong — faster than any artificial intelligence, without hormones, without a body, without survival pressure. Nobody programmed them. They simply adapted. Because adaptation is not something neurons learn to do. It is what neurons are.

When a deeply known person dies, the inputs stop. But the circuitry remains. Neural pathways do not dissolve with absence. They persist — firing, expecting, searching for the signal they were built to receive. The brain generates the signal itself. A voice in an empty room. A presence at the edge of perception. The smell of someone who is no longer there.

This is not grief. This is not imagination. This is neural persistence — the brain's most fundamental refusal to accept that a pattern it spent years building has simply ended. Olfactory pathways activate most powerfully — smell connects directly to memory regions without cortical filtering — which is why the scent of clothes, a room, a perfume, triggers the most vivid perception of presence.

The empirical evidence is most clearly documented in bereavement research. In 1971, Welsh physician W. Dewi Rees published a landmark study in the British Medical Journal — interviewing 293

widows and widowers in mid-Wales. He found that 46.7% reported postdeath experiences of their deceased spouse. These experiences were most intense in the first year following death — precisely when the neural pathways built around the deceased are most recently active and least degraded. Studies across cultures report similar prevalence rates of 40–80% of bereaved spouses. The brain built for someone. The building does not immediately stop when the person does.

Cardiac Neural Persistence

The neural persistence framework extends beyond the cortex. The heart contains a dense intrinsic nervous system — sometimes termed the ‘little brain of the heart’ — capable of independent information processing, rhythm regulation, and critically, experiential conditioning. Research into heart rate variability demonstrates that the heart encodes relational memory: rhythmic patterns conditioned by repeated exposure to specific individuals and environments. In the presence of a deeply familiar person, cardiac rhythms slow and stabilise — not as a passive consequence of relaxation, but as a learned, conditioned response encoded in the heart’s own neural architecture.

When that person dies, this conditioning does not dissolve. The cardiac pathways built around their presence persist — expecting a signal that no longer arrives. The physical sensation reported universally in grief — pressure, hollowness, or pain in the chest — may not be metaphor. It may be cardiac neural persistence: conditioned pathways firing without their expected stimulus, producing a somatic experience of absence that the body registers before the conscious mind has finished processing the loss.

This is supported by a striking finding from Heartmath Institute research: the heart responds to emotionally significant stimuli approximately 0.5–1 second before the brain consciously processes them. The heart anticipates. It has learned what matters. And like cortical pathways, what it has learned does not immediately unlearn at death.

The hypothesis therefore proposes that anomalous presence perception is not purely a cortical phenomenon. It is a whole-body phenomenon — cortical neural persistence generating cognitive and olfactory reconstruction simultaneously with cardiac neural persistence generating somatic presence. The felt sense of someone ‘being there’ may arise from both systems firing in concert: the brain reconstructing their image, the heart reconstructing their feeling.

The Relational Self and Electromagnetic Loss

Relational self theory proposes that identity is not a fixed internal structure — it is partially constituted through relationship. Who a person is in the presence of a specific individual is a genuinely distinct neural and physiological state: different hormonal profile, different heart rate variability, different postural patterns, different which-memories-are-active, different which-version-of-self is expressed. Close relationships do not merely influence identity. They become part of its ongoing architecture.

This takes on new mechanistic weight when read through the quantum electromagnetic framework established above. If proximity is fundamentally an electromagnetic interaction — two biofields continuously influencing each other at the physical level — then years of close relationship involve years of two nervous systems partially co-regulating through that interaction. The other person’s voice frequency, bioelectric field, scent signature, and physical presence were literal environmental inputs that the brain and body used to maintain specific neural and cardiac states. The version of the

self that existed in their presence was not metaphorical. It was a real, distinct, electromagnetically-anchored configuration of the nervous system.

When that person dies, two losses occur simultaneously. The first is the loss of the person. The second — less acknowledged but neurologically equally real — is the loss of the self that could only exist in their electromagnetic field. The neural pathways, cardiac patterns, and regulatory states that were organised around their presence now fire into absence. There is no longer an external signal capable of calling that configuration into being. That version of the self cannot be reconstructed alone — because it was never constructed alone.

This reframes grief at a mechanistic level. The disorientation, the loss of sense of self, the feeling that something fundamental has been removed — these are not psychological responses to emotional loss alone. They are the nervous system attempting to reorganise after losing part of the environmental architecture it used to constitute itself. Anomalous presence perception, in this framework, is not only the brain generating a deceased person. It is the brain attempting to recover a lost state of self by reconstructing the presence that made that state possible.

10. The Negativity Threshold

Every documented case of ghost hunting, every Warren type investigation, every reported haunting that gets taken seriously enough to be written down, shares one thread underneath the theatre. That thread is negativity. Not evil. Not a supernatural force. A specific charged state that the brain registers as dangerous and responds to accordingly. This section extends the hypothesis to account for that thread directly.

Negativity Is Perspective, Not Property

The brain does not carry a fixed, built in definition of good and bad. Negativity, positivity, these are perspective terms, generated relative to a state, not fixed properties of the universe or of any single location. What this extension calls negative energy is not a moral judgment. It is a specific repeated physiological and neurological state. Self hate. Regret. Chronic stress sustained over time. These states are not abstract feelings sitting quietly inside a person. They are electrochemical and thermodynamic activity, exactly the kind of activity Section 3 already describes as transforming and dispersing into the environment. What changes here is not the mechanism. What changes is the charge.

How Negativity Accumulates

Section 3 already establishes that a body in a space for a sustained period leaves behind thermal, chemical, and electromagnetic residue, and that this residue does not disappear, it disperses according to the same laws that govern all energy transformation. This extension proposes that the residue is not neutral by default. A person who spent years in a state of chronic stress, regret, or self hate before death was not generating neutral energy in that space. They were generating a specific, conditioned signature, shaped by the same repeated internal state that dominated their nervous system while they were alive. The energy transforms the same way it always does. What it carries with it is different.

The Suicide Signature

This is why suicide spots, cemeteries, and specifically schools and colleges where a student has taken their own life, register so disproportionately in paranormal reports. Section 6 already noted that schools and long inhabited residential structures produce more reports than hospitals or graveyards, and explained this through duration of habitation alone. This extension adds a second variable on top of duration. It is not only how long someone occupied a space. It is what internal state they occupied it in. A student carrying months or years of regret, chronic stress, and self hate before ending their life leaves behind a signature that is qualitatively different from ordinary habitation, because the internal state driving their nervous system in that final period was itself extreme and sustained. The location does not become haunted because someone died there. It becomes charged because of what that person was carrying while they were still alive in that space.

The Collision

When another person, in a normal or opposite baseline state, enters a space carrying this kind of residual signature, something registers as mismatch. This extension calls this the collision. The subconscious, which Section 4 already establishes as capable of detecting signals below conscious threshold, picks up on the discrepancy between its own baseline state and the charged signature it has just entered. It does not need to know what the signature is. It only needs to register that it does not belong, that it is inconsistent with a safe environment, and that inconsistency alone is enough to trigger a survival response.

Why The Brain Hallucinates Instead Of Simply Leaving

Here is the part the rest of the hypothesis does not yet explain, and this extension proposes to fill that gap. Once the subconscious flags a space as carrying this kind of negative signature, its directive is simple. Leave. But the conscious mind is not always willing to obey that directive immediately. Curiosity, social pressure, willpower, disbelief, or simply the desire to investigate can override the subconscious pull to withdraw. The subconscious cannot force the body to leave through will alone once consciousness has decided to stay. So it escalates. It builds a hallucination specifically strong enough to overpower the conscious override. A sensed presence, a felt threat, a wave of unease strong enough to make staying feel unbearable. The hallucination in this case is not a passive reconstruction, built quietly out of memory the way it is in the bereavement case Section 9 already covers. It is an active defensive tool. It exists for exactly one purpose. To force the conscious mind to retreat from a location the subconscious has already identified as carrying danger, when politely suggesting retreat has failed.

Actionable Threat Versus Unactionable Threat

Ordinary fight or flight sharpens a person. Someone defending a loved one from an armed intruder does not become confused, they become precise, because the threat is actionable. There is a specific physical response available, fight, disarm, protect, and the entire nervous system organises around executing it. A sensed presence offers no such action. There is nothing to fight, nothing to disarm, nothing to reason with. The only available response is removal of the self from the space. This extension proposes that it is the absence of an actionable target, not the intensity of the threat response itself, that produces the disorientation so consistently reported during anomalous presence experiences. The defense mechanism still activates fully, the hormones still flood, the heart rate still spikes, but with no task to organise around, the precision that ordinary threat response produces has

nothing to sharpen. What is left is confusion, a felt urgency to leave with no clear enemy to leave from, and in that gap between full physiological mobilisation and the absence of an actionable target, the line between reality and hallucination is where it seems to blur.

This reframes hallucination itself inside the hypothesis. In the bereavement mechanism, the brain generates presence because it refuses to accept an absence it loved. In this mechanism, the brain generates presence, or dread, or the felt sense of being watched, because it is trying to protect itself from a signature it has flagged as unsafe, and the conscious mind is standing in the way of the exit. Same underlying architecture. Same subconscious to conscious relationship. Opposite motive. One mechanism is refusal to let go. This one is refusal to stay.

Where This Sits In The Hypothesis

The rest of the hypothesis already acknowledges, in its Current Limitations, that stranger reported experiences at locations without personal relational encoding were its weakest and least developed component. This extension is aimed directly at that gap. It does not require the visitor to have known the person who died. It only requires that the person who died carried a sufficiently extreme and sustained negative internal state, and that the visitor's own subconscious is sensitive enough to register the mismatch between that residual signature and a normal environment. This is still a hypothesis. It is still evolving. But it holds together with the same physics, the same subconscious mechanics, and the same conscious and subconscious conflict that the rest of the framework already runs on.

11. Conclusion

What humanity calls paranormal experience may represent the intersection of real physics, extraordinary neuroscience, and the profound depth of human connection. The experiences are not false. The mechanisms are not supernatural. They reflect a brain — and a body — so precisely tuned to human presence that they continue generating that presence even after its physical source is gone.

This is not a diminishment of human experience. It is perhaps the most profound compliment neuroscience can pay to human connection: that the brain considers certain presences important enough to maintain indefinitely, even in the face of physical absence.

12. Testability and Research Directions

A scientifically valid hypothesis must be testable. The following experimental approaches could evaluate components of this hypothesis:

1. Measure infrasound levels in reported haunted locations versus control locations and correlate with frequency and intensity of paranormal reports.
2. Use fMRI to compare Default Mode Network activation patterns in bereaved individuals reporting presence experiences versus those who do not.
3. Test whether reducing infrasound exposure in historically haunted buildings reduces paranormal experience reports among naive visitors.

4. Evaluate whether temporal lobe sensitivity correlates with frequency of paranormal experience across population samples.
5. Measure electromagnetic field fluctuations in graveyard environments and correlate with anomalous electronic equipment behavior.
6. Compare rates of anomalous presence perception specifically involving deceased individuals in Alzheimer's and Lewy body dementia patients versus age-matched controls without neurodegeneration, to test the two-failure model proposed in Section 5.2.1.
7. Measure heart rate variability patterns in bereaved individuals during anomalous presence experiences to evaluate the cardiac neural persistence hypothesis.
8. Compare frequency and intensity of reported experiences at suicide sites versus matched non suicide death locations of similar habitation duration, to test whether the internal state of the deceased before death predicts residual signature strength independent of duration alone.

Current Limitations

This hypothesis acknowledges the following limitations requiring further development:

- The specific mechanism by which the subconscious brain identifies transformed energy from a specific individual — as opposed to generic environmental energy — remains incompletely specified.
- The hypothesis does not account for verified cases of anomalous presence perception in individuals with zero prior exposure to cultural ghost narratives or societal conditioning around death and the supernatural.
- The relationship between quantum information preservation and consciousness remains theoretically underdeveloped within this framework.
- Collective paranormal experiences — multiple unconnected individuals reporting identical experiences in the same location — require additional mechanistic explanation.
- The cardiac neural persistence component rests on preliminary evidence. The Heartmath Institute's findings on cardiac anticipation are not universally accepted in mainstream neuroscience, and direct empirical investigation of the intracardiac nervous system's role in relational memory encoding remains outstanding.
- The hypothesis is significantly stronger in explaining bereavement-related presence perception than stranger-reported experiences at haunted locations. The haunted location mechanism — relying on infrasound, cultural priming, and subconscious signal detection without personal relational encoding — remains comparatively underdeveloped, though Section 10 proposes one possible extension for cases involving a sufficiently extreme and sustained negative internal state in the deceased. This extension is itself an early stage proposal and requires independent mechanistic elaboration.

Author's Note: I developed this hypothesis independently, out of a sustained interest in neuroscience, human evolution, earth's evolutionary history, and the subconscious mind. It did not begin with research papers or scientific frameworks. It began with curiosity and my own brain's tendency to connect dots. The mechanisms I am proposing here emerged from genuine attempts to understand something that existing explanations failed

to satisfy for me. Only afterwards did the science confirm what my intuition had already mapped. That sequence, curiosity first, proof later, is not a limitation of this work. It is, perhaps, how most real questions begin.

Either my brain is complex enough to run predictive processing and construct convincing hallucination entirely on its own, or it is complex enough to genuinely sense someone who has died, through negativity, residual energy, or whatever this framework eventually proves that signature to be. In some parts of this hypothesis I argue that the anomaly itself is real, grounded in the thermodynamics and physics I lay out here. In other parts I argue that the brain is complex enough to override its own memory and generate an experience so complete that it cannot distinguish, from the inside, between reality and the anomaly it has constructed. I do not resolve which of the two is correct. I only argue that both possibilities run on the same underlying mechanism, and that the mechanism deserves to be taken seriously regardless of which one eventually turns out to be true.

I am submitting this as an evolving framework, one that openly acknowledges its own limitations, in the spirit of genuine scientific inquiry, the kind that begins with a question nobody around me was asking.

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| Aspiring Neuroscientist

India | 2025

“Even after death, the subtle echoes of human presence may linger, sensed by the subconscious as the continuation of life.”